Optimizing Sales, Inventory and Margin For Eesbee Garments

A Proposal report for the BDM capstone Project

Submitted by

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1. Executive Summary

Eesbee is a B2C single-door garment store located in Alkapuri, Hyderabad. It offers branded innerwear, kidswear, and ethnic garments from Jockey, US Polo, Van Heusen, Ramraj and others, catering to walk-in customers from nearby areas.

Eesbee is a profitable business that is owned by two partners. However, it faces operational challenges related to inventory inefficiencies, margin variability, and limited visibility into product-level performance. Currently, there is no structured analysis of stock movement, bundling opportunities, or pricing effectiveness. The business uses separate systems for billing and expenses. It doesn't capture customer data, have a marketing strategy, or use any analytical tools. Inventory entries are manually recorded from paper invoices, and sales are captured via barcode scanning. The store operates three desktop computers, and data is stored locally in SQL Server database.

This project will analyze sales, purchase, and inventory data from April 1st 2023 to September 30th 2025 to uncover inefficiencies and support data-driven decisions. A modular Python program will be developed to perform margin diagnostics, inventory movement classification, and sales basket modeling. It will also document operational constraints, like batch identification challenges during billing, and suggest alternatives. The expected outcome is a set of actionable insights that reduce inventory blockage, improve pricing and stocking decisions, and enhance profitability through better operational control.

2. Organization Background

Eesbee operates from a 1000 SFT rented space on a busy street. It has separate sections for men's and women's apparel. The store turnover is around 2 crore per annum and it has approximately 21,000 items from around 60 suppliers. Inventory is usually provided on credit for three months, and there are occasional consignment arrangements for new launches.

Operations are managed by two owners and eight employees, covering sales, billing, accounting, and store management. All sales are walk-in, and customer data isn't captured. The store does not run promotions, loyalty programs, or online sales channels.

Billing and inventory are handled via Quick Bill POS, while expenses and GST filings are managed by Tally ERP. These systems are standalone and are not integrated. Data is stored locally

on a SQL Server and exported in CSV format for analysis. A vendor support contract ensures periodic maintenance of the POS system.

3. Problem Statement

Eesbee's current systems capture sales and inventory data, but lack analytical depth. Key gaps include margin visibility, inventory movement classification, and sales basket analysis. These limitations hinder data-driven decision-making and reduce operational efficiency. The project aims to explore these gaps through structured diagnostics and modular analysis. Additionally, a critical operational pain point—batch identification during billing—has emerged as a central concern.

3.1 Margin Visibility

There is no SKU-level tracking of margin trends, making it difficult to identify low-margin items or prioritize profitable categories.

3.2 Inventory movement

Stock velocity is not monitored, and items are not classified by movement type, limiting reorder planning and shelf optimization.

3.3 Sales Basket Patterns

Customer purchase combinations are not analyzed, missing opportunities for bundling, cross-selling, and improved product placement.

3.4 Batch Identification Challenge

Items cannot be traced by incoming batch during billing, as brand barcodes lack batch-level data. This limits post-sale traceability and complicates customer queries, especially when resolving complaints or verifying purchase history.

4. Background of the Problem

Eesbee's operational setup presents many analytical gaps that limit decision-making based on data. The store captures sales and inventory data through its POS system, but there is no structured analysis of profitability, inventory movement, or customer purchase behavior. These gaps come from both internal and external limitations.

Internally, there is an absence of SKU-level margin tracking making it difficult to monitor profitability trends or identify low-margin items. The movement of inventory is not classified, reducing identification of fast-moving, slow-moving, or dead stock. Sales data is available at the item level, but customer purchase combinations are not analyzed, missing opportunities for

bundling and cross-selling. Additionally, billing doesn't support batch-level tracing, which complicates post-sale analysis.

Externally, the supplier terms vary across different credit and consignment models, and seasonal product launches introduce variability in stock performance. Without analytical tools, the store cannot evaluate these patterns or adjust future stocking strategies effectively.

This project applies Business Data Management (BDM) concepts to address these gaps. Modular Python tools will be implemented for Margin diagnostics, inventory classification, and basket analysis. Operational constraints will be documented to reflect real-world traceability challenges and support future system improvements.

5. Problem Solving Approach

This project applies Business Data Management (BDM) principles to uncover inefficiencies in Eesbee's retail operations. The approach combines structured data collection, targeted analytical methods, and modular tooling to address margin variability, inventory inefficiencies, and missed bundling opportunities.

5.1 Analytical Methods and Rationale

Margin Diagnostics: Sales and purchase data of items will be aggregated to compute margins over time. This allows identification of low-margin SKUs, margin erosion trends, and pricing anomalies. ***Aggregation and time-series techniques*** will support profitability analysis.

Inventory Movement Classification: Purchase, return, and sales data will be used to estimate stock turnover and classify items into fast-moving, slow-moving, and dead stock. This informs how to plan for reordering and optimize shelf space. Items will be classified based on ***velocity thresholds and movement frequency***.

Sales Basket Modeling: Transactional data will be examined by applying association rule mining to identify frequently co-purchased items. This informs bundling strategies, product placement, and cross-selling opportunities. ***Apriori algorithm*** will be used to generate itemset rules.

Operational Insight: Batch identification challenges will be documented through process mapping. While this is not analytically modeled, it reflects real-world limitations associated with traceability and informs future system design.

5.2 Intended Data Collection

Data will be collected from Eesbee's POS and accounting systems for the period ***April 1st, 2023*** to ***September 30th, 2025***. This timeframe follows a system crash and a manual stock reset, ensuring consistency. Four datasets will be used:

Inventory Data: This dataset includes opening/closing balances, supplier-wise purchases, and item return logs. It will be used for margin and movement analysis.

Sales Data: It consists of item-wise transactions with timestamps, quantity sold, and selling price. It will support margin diagnostics and basket modeling, but limiting to item-level associations.

Expense Summary: It contains the monthly operating expenses, used to estimate breakeven points and margin-to-cost trends.

Physical Stock Verification: Pending; It will be used to validate system-recorded inventory and identify discrepancies.

5.3 Analysis Tools

All modules will be developed in **Python** with plug and play architecture in mind. **Pandas** will be used for data wrangling, **Matplotlib/Seaborn** for visualization, and **MLxtend** for association rule mining. All input data will be in **CSV** format.

5.4 Scope Limitations

This study does not include customer-level analysis, as demographic and loyalty data is not available. The store is offline only, means online sales and digital marketing are not included. The Batch-level traceability is documented since it is a customer requirement, but it is not analytically modeled. Expense data is available only in aggregated form by month, which limits granularity in cost attribution. System integration between POS and accounting platforms hasn't been done and is outside the scope of this project. Report of physical stock verification is still pending and may affect accuracy of inventory movement analysis.

6. Expected Timeline

6.1 Work Breakdown Structure

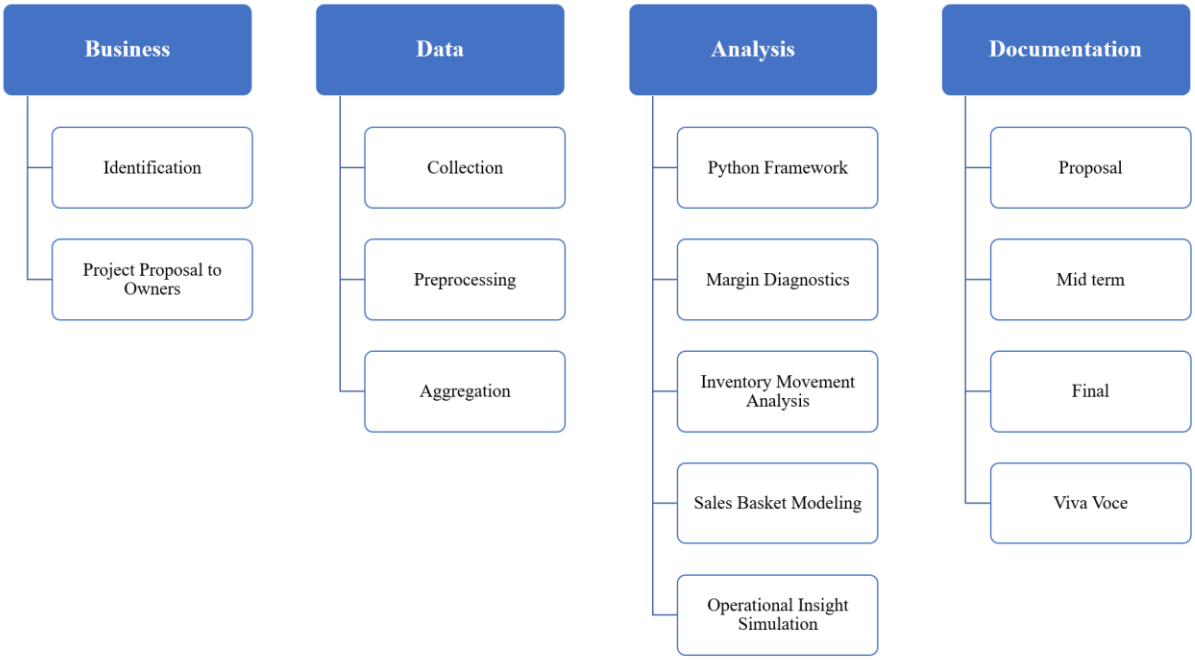

Figure 1: Work Breakdown Structure

6.2 Gantt Chart

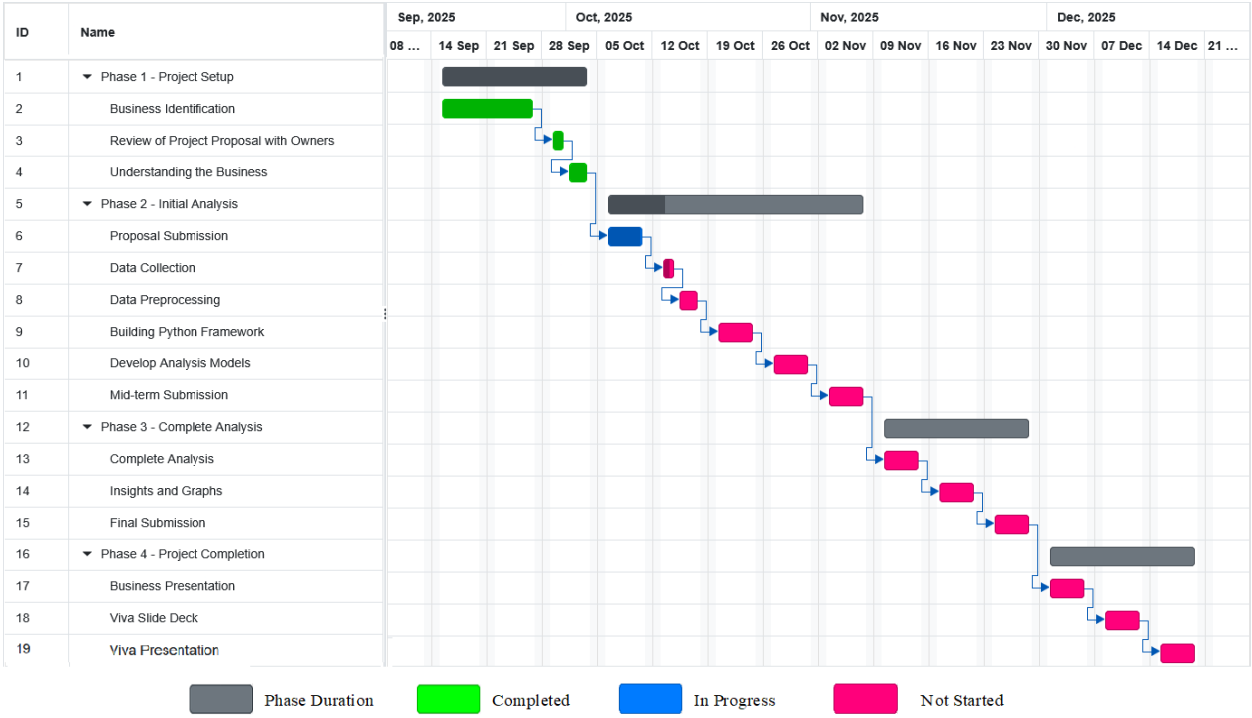

Figure 2: Gantt Chart for the BDM Capstone Project

7. Expected Outcome

This project aims to address key operational gaps at Eesbee through structured analysis of sales, inventory, and supplier data. Each objective is designed to create actionable insights that can be implemented for decision-making and operational efficiency.

- **Margin Diagnostics**: Analyze trends in SKU-level margins using purchase and sales data, to identify products with consistent margin erosion or gain. Classify SKUs by profitability to support pricing decisions and prioritizing inventory.
- **Inventory Movement Analysis**: Estimate stock turnover and classify items as fast-moving, slow-moving, or dead stock using supplier records, sales transactions, and return logs. Identify inefficiencies and support reorder planning and shelf space optimization.
- **Sales Basket Patterns**: Apply association rule mining to item-wise sales data to identify products frequently purchased together. Generate item combinations that inform bundling strategies, product placement, and cross-selling opportunities.
- **Operational Insight (Non-Analytical)**: Document the batch identification challenge during billing, which affects post-sale traceability. While this is not part of the analytical scope, this operational constraint will be simulated to reflect real-world limitations and the context that initiated the project.

The outcomes will be delivered through modular Python tools and structured documentation.